



Module Presenter's Manual *for* **Application Development using Flutter and Dart**

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Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	June 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Learn to code Flutter apps using Dart programming language
- Use Flutter framework and Dart programming language to develop cross platform mobile apps
- Learn platform-specific Widgets, platform independent Widgets, Single Child Layout Widgets, and Multiple Child Layout Widgets
- Explore state management, dependency injection, navigation, routing, and controllers in Flutter
- Learn database concepts with Sqflite and Firebase and REST API

2. Information on Session Allocation

Module	Online (No. of Hrs)
Application Development using Flutter and Dart	40

Throughout this Presenter's Manual, the module Application Development using Flutter and Dart will be referred to as **FLDA**.

3. Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration

➤ **Week-Wise Schedule**

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 FLDA-TL1	Session 2 FLDA-TL2	Session 3 FLDA-TL3	Session 4 FLDA-TL4
2	Session 5 FLDA-TL5	Session 6 FLDA-TL6	Session 7 FLDA-TL7	Session 8 FLDA-TL8
3	Session 9 FLDA-TL9	Session 10 FLDA-TL10	Session 11 FLDA-TL11	Session 12 FLDA-TL12
4	Session 13 FLDA-TL13	Session 14 FLDA-TL14	Session 15 FLDA-TL15	Session 16 FLDA-TL16
5	Session 17 FLDA-TL17	Session 18 FLDA-TL18	Session 19 FLDA-TL19	Session 20 FLDA-TL20

TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	FLDA- TL1	All the topics as listed below from Session 1 of Application Development using Flutter and Dart book should be covered in this session. <u>Session 1 – Introduction to Flutter</u> ➤ Define Flutter framework ➤ List and describe features of Flutter ➤ List the advantages of using Flutter ➤ Outline the Flutter architecture ➤ Identify components of Flutter ➤ Explain installation of Flutter in Android Studio	<u>Application Development using Flutter and Dart</u> SG - Session 1 XP - Session 1 TG- Session 1
2	FLDA- TL2	All the topics as listed below from Session 2 of Application Development using Flutter and Dart book should be covered in this session. <u>Session 2 – Applications and Widgets in Flutter</u> ➤ Explain how to create a simple Flutter application in Android Studio ➤ Explain Widgets in Flutter and uses ➤ List different types of widgets in Flutter ➤ List different types of layout Widgets	<u>Application Development using Flutter and Dart</u> SG - Session 2 XP - Session 2 TG- Session 2
3	FLDA- TL3	The Try It Yourself questions of Session 1 and Session 2 of <i>Application Development using Flutter</i> book should be covered in this session.	<u>Application Development using Flutter and Dart</u> Session 1 & 2
4	FLDA- TL4	All the topics as listed below from Session 3 of Application Development using Flutter and Dart book should be covered in this session. <u>Session 3 – Building Flutter Application Using Platform Specific Widgets</u> ➤ Explain Scaffold and AppBar ➤ Describe BottomNavigationBar	<u>Application Development using Flutter and Dart</u> SG - Session 3 XP - Session 3 TG- Session 3

Sessi on No.	Session Title	Session Details	Deliverables' Mapping
		➤ Describe TabBar and TabBarView ➤ Describe ListTile ➤ Identify different types of Buttons and explain them ➤ Define User Input Widgets ➤ Explain DatePickers and Dialogs	
5	FLDA- TL5	All the topics as listed below from Session 4 of Application Development using Flutter and Dart book should be covered in this session. <u>Session 4 – Building Flutter Application Using Platform Independent Widgets</u> ➤ Explain Text widget ➤ Explain Image widget ➤ Explain Icon widget ➤ Explain Form widget	<u>Application Development using Flutter and Dart</u> SG - Session 4 XP - Session 4 TG- Session 4
6	FLDA- TL6	The Try It Yourself questions of Session 3 and Session 4 of <i>Application Development using Flutter</i> book should be covered in this session.	<u>Application Development using Flutter and Dart</u> Session 3 & 4
7	FLDA- TL7	All the topics as listed below from Session 5 of Application Development using Flutter and Dart book should be covered in this session. <u>Session 5 – Building a Flutter Application Using Single Child Layout Widgets</u> ➤ Describe Single Child Layout widgets ➤ Explain Container widget, Padding widget, Align widget, Center widget, and SizedBox widget	<u>Application Development using Flutter and Dart</u> SG - Session 5 XP - Session 5 TG- Session 5
8	FLDA- TL8	All the topics as listed below from Session 6 of Application Development using Flutter and Dart book should be covered in this session. <u>Session 6 – Building a Flutter Application Using Multi Child Layout Widgets</u> ➤ Explain Multi Child Layout widgets ➤ Define and explain Row widget, Column widget, Stack widget, and other child	<u>Application Development using Flutter and Dart</u> SG - Session 6 XP - Session 6 TG- Session 6

Sessi on No.	Session Title	Session Details	Deliverables' Mapping
		layout widgets	
9	FLDA- TL9	The Try It Yourself questions of Session 5 and Session 6 of <i>Application Development using Flutter</i> book should be covered in this session.	<u>Application Development using Flutter and Dart</u> Session 5 & 6
10	FLDA- TL10	All the topics as listed below from Session 7 of Application Development using Flutter and Dart book should be covered in this session. <u>Session 7 – State Management and Dependency Injection in Flutter</u> ➤ Describe State Management in Flutter ➤ Explain State Management types in Flutter ➤ Define Dependency Injection in Flutter - Outline example of dependency injection in Flutter	<u>Application Development using Flutter and Dart</u> SG - Session 7 XP - Session 7 TG- Session 7
11	FLDA- TL11	All the topics as listed below from Session 8 of Application Development using Flutter and Dart book should be covered in this session. <u>Session 8 – Navigation, Routing, and Controllers in Flutter</u> ➤ Define Navigation and Routing ➤ Outline Routes in Flutter ➤ Explain Routing class ➤ Describe navigation concept in Flutter ➤ Explain Controllers in Flutter ➤ Describe working with Controllers in Flutter ➤ Define Custom controller in Flutter	<u>Application Development using Flutter and Dart</u> SG - Session 8 XP - Session 8 TG- Session 8
12	FLDA- TL12	The Try It Yourself questions of Session 7 and Session 8 of <i>Application Development using Flutter</i> book should be covered in this session.	<u>Application Development using Flutter and Dart</u> Session 7 & 8
13	FLDA- TL13	All the topics as listed below from Session 9 of Application Development using Flutter and Dart book should be covered in this session. <u>Session 9 – Styling and Creating Responsive Applications in Flutter</u>	<u>Application Development using Flutter and Dart</u> SG - Session 9 XP - Session 9 TG- Session 9

Sessi on No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain working of themes in Flutter ➤ Outline the Styling of most used Widgets: Text, Container, and RaisedButton ➤ Illustrate Toggling between themes ➤ Explain responsive Flutter applications ➤ Describe MediaQuery.of() ➤ Explain LayoutBuilder and other widgets used for responsiveness	
14	FLDA- TL14	The Try It Yourself questions of Session 9 of <i>Application Development using Flutter</i> book should be covered in this session.	<u>Application Development using Flutter and Dart</u> Session 9
15	FLDA- TL15	All the topics as listed below from Session 10 of <i>Application Development using Flutter and Dart</i> book should be covered in this session. <u>Session 10 – FutureBuilder and StreamBuilder in Flutter</u> ➤ Explain Future and FutureBuilder ➤ Explain Stream and StreamBuilder ➤ List the differences between FutureBuilder and StreamBuilder	<u>Application Development using Flutter and Dart</u> SG - Session 10 XP - Session 10 TG- Session 10
16	FLDA- TL16	The Try It Yourself questions of Session 10 of <i>Application Development using Flutter</i> book should be covered in this session.	<u>Application Development using Flutter and Dart</u> Session 10
17	FLDA- TL17	All the topics as listed below from Session 11 of <i>Application Development using Flutter and Dart</i> book should be covered in this session. <u>Session 11 – Database Concepts with Sqflite and Firebase Database</u> ➤ Describe Sqflite in Flutter and its usage ➤ Elaborate the process of installing SQLite in Flutter ➤ Explain executing SQLite Queries in Flutter ➤ Explain working with Firebase Database in Flutter	<u>Application Development using Flutter and Dart</u> SG - Session 11 XP - Session 11 TG- Session 11
18	FLDA- TL18	The Try It Yourself questions of Session 11	<u>Application</u>

Sessi on No.	Session Title	Session Details	Deliverables' Mapping
		of <i>Application Development using Flutter</i> book should be covered in this session.	<u>Development using Flutter and Dart</u> Session 11
19	FLDA- TL19	All the topics as listed below from Session 12 of <i>Application Development using Flutter and Dart</i> book should be covered in this session. <u>Session 12 – REST API in Flutter</u> ➤ Explain REST APIs ➤ Define and explain utilizing APIs for REST in Flutter ➤ Explain the use of http package ➤ Outline the core methods of the HTTP component ➤ Explain the use of Flutter apps to showcase REST APIs ➤ Explain how to register and log in utilizing sample Web APIs	<u>Application Development using Flutter and Dart</u> SG - Session 12 XP – Session 12 TG- Session 12
20	FLDA- TL20	The Try It Yourself questions of Session 12 of <i>Application Development using Flutter</i> book should be covered in this session.	<u>Application Development using Flutter and Dart</u> Session 12

6. Library References

➤	Flutter Complete Reference by Alberto Miola
➤	Flutter for Beginners: An introductory guide to building cross-platform mobile applications with Flutter 2.5 and Dart, 2nd Edition 2nd ed. Edition by Thomas Bailey, Alessandro Biessek, Trevor Wills
➤	Real-World Flutter by Tutorials (First Edition): Professional App Development With Flutter by raywenderlich Tutorial Team, Edson Bueno, Vid Palcar

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